ACS Sustainable Chemistry & Engineering

Research Article

Cite This: ACS Sustainable Chem. Eng. 2019, 7, 9211−9219

pubs.acs.org/journal/ascecg

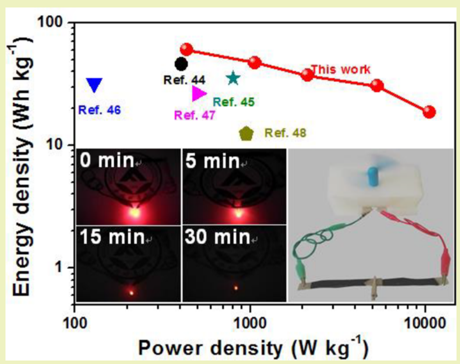
Fe_2O_3 Nanoparticles Decorated on Graphene-Carbon Nanotubes Conductive Networks for Boosting the Energy Density of All-Solid-State Asymmetric Supercapacitor

Yadong Tian, Xin Hu, Yahui Wang, Chen Li, and Xiaoliang Wu*

Department of Chemistry and Chemical Engineering, College of Science, Northeast Forestry University, 26 Hexing Road, Harbin 150040, P. R. China

Supporting Information

ABSTRACT: Ferric oxide (Fe_2O_3) has drawn massive attention as a promising cathode electrode material for supercapacitors because of its large theoretical specific capacity, low cost, and abundance in nature. Nevertheless, its relatively low conductivity and large volume change seriously impede its electrochemical performance. Herein, Fe_2O_3 nanoparticles decorated on graphene-carbon nanotubes (CNTs) conductive networks (Fe_2O_3/GNs/CNTs) were prepared by a simple reflux way. Owing to the unique structure, the Fe_2O_3/GNs/CNTs electrode delivers a notably enhanced specific capacity (675.7 F g^{-1} at 1 A g^{-1}) and superior rate characteristic in 6 M KOH aqueous electrolyte. More importantly, the as-constructed all-solid-state asymmetric device using Fe_2O_3/GNs/CNTs as the cathode electrode and sulfurized CoAl layered double hydroxides (SLDH) as the anode electrode shows the high energy density of 60.3 Wh kg^{-1} and good electrochemical steadiness in KOH/PVA gel electrolyte. Therefore, this strategy provides a novel mothed to synthesize Fe_2O_3 based electrode materials for energy storage system.



KEYWORDS: *Fe_2O_3, Graphene, CNTs, Energy density, Asymmetric supercapacitor*

INTRODUCTION

As resources wilt and environmental damage becomes more and more serious, it is imperative to develop environmentally friendly, efficient, and inexpensive facilities for energy storage.[1−3] Among various energy storage facilities, the supercapacitor has attracted tremendous interests owing to the inimitable features, for instance ultrahigh power density, ultrafast charge−discharge characteristic, superior electrochemical stabilization and environment friendly.[4−7] Nevertheless, most of supercapacitors are confronted with a relatively low energy density, which seriously restricts their widely utilization.[8] Therefore, it is urgent to enhance the energy density of supercapacitors to satisfy the increasing demand for high energy density facilities while maintaining their large power density. Based on the formula $E = 1/2CV^2$, the enhancement of energy density of supercapacitors can be obtained through improve the specific capacity (C) or/and enlarge the potential range (V).

Assembling asymmetric supercapacitors (ASCs) have proved to be an efficacious way to enlarge the device voltage range by a combination of the voltage range of cathode and anode electrodes, resulting in a remarkable improvement of energy density.[9−12] Up to now, a significant advancement has been made to develop anode electrodes for ASCs, for example MnO_2, Co_3O_4, $Ni(OH)_2$, $NiCoS_4$, etc.[13−18] Porous carbon with a double layer charge storage mechanism is usually used as cathode electrodes, which usually suffer from low specific capacity. Consequently, the energy density of ASCs is badly limited by the poor specific capacity of cathode electrodes. As a kind of cathode electrode material, ferric oxide (Fe_2O_3) materials have received extensive attentions because of the large theoretical specific capacity, low cost and abundance in nature.[19−24] Especially, iron element owns various oxidation states, which can generate reversible redox reaction within the cathode voltage regions. Such as, Shivakumara synthesized porous flower-like α-Fe_2O_3 material through the self-assembly strategy and the prepared electrode delivers a capacitance of 127 F g^{-1} at 1 A g^{-1}.[25] Zheng prepared hollow nanoshuttle-like α-Fe_2O_3 through a convenient hydrothermal way, and the obtained materials deliver a large capacity of 249 F g^{-1} at 0.5 A g^{-1}.[26] Although, great achievements have been made, the electrochemical properties of Fe_2O_3 are not satisfied with actual applications because of the low conductivity and large volume change. Currently, combining with conductive materials to construct composite is an efficacious method to improve the conductivity and capacitive property of Fe_2O_3-based materials.[20] Nanostructured carbon materials for instance CNTs and graphene are regarded as an excellent conductive substrate for construction of composites because of the good conductivity and high specific surface area.[27,28]

Received: December 30, 2018
Revised: April 3, 2019
Published: April 27, 2019

DOI: 10.1021/acssuschemeng.8b06857
ACS Sustainable Chem. Eng. 2019, 7, 9211−9219

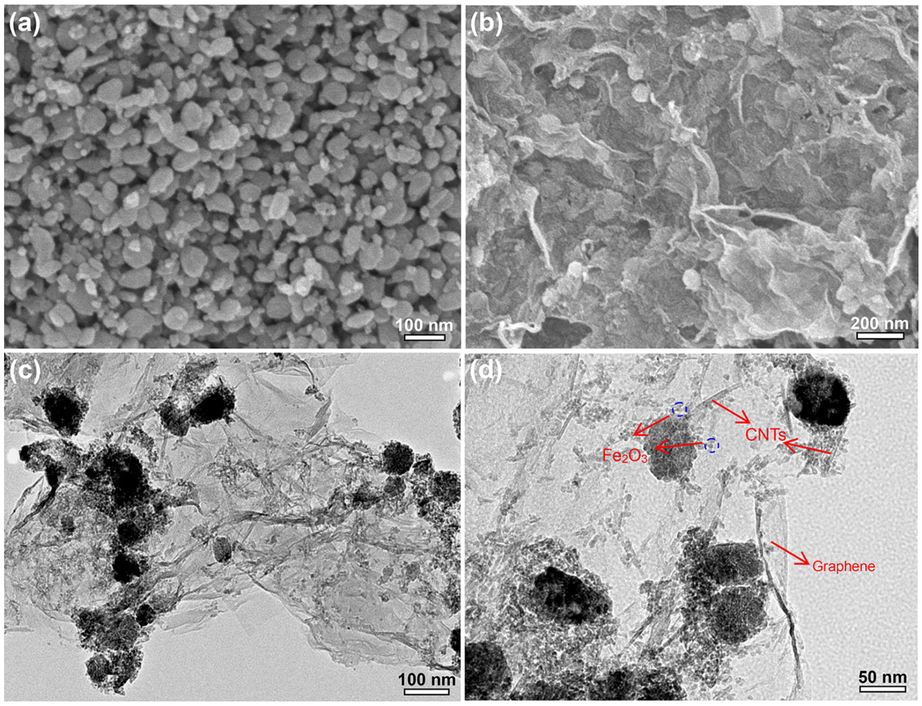


Figure 1. SEM images of (a) pure Fe_2O_3 and (b) Fe_2O_3/GNs/CNTs. (c, d) TEM images of the Fe_2O_3/GNs/CNTs composite.

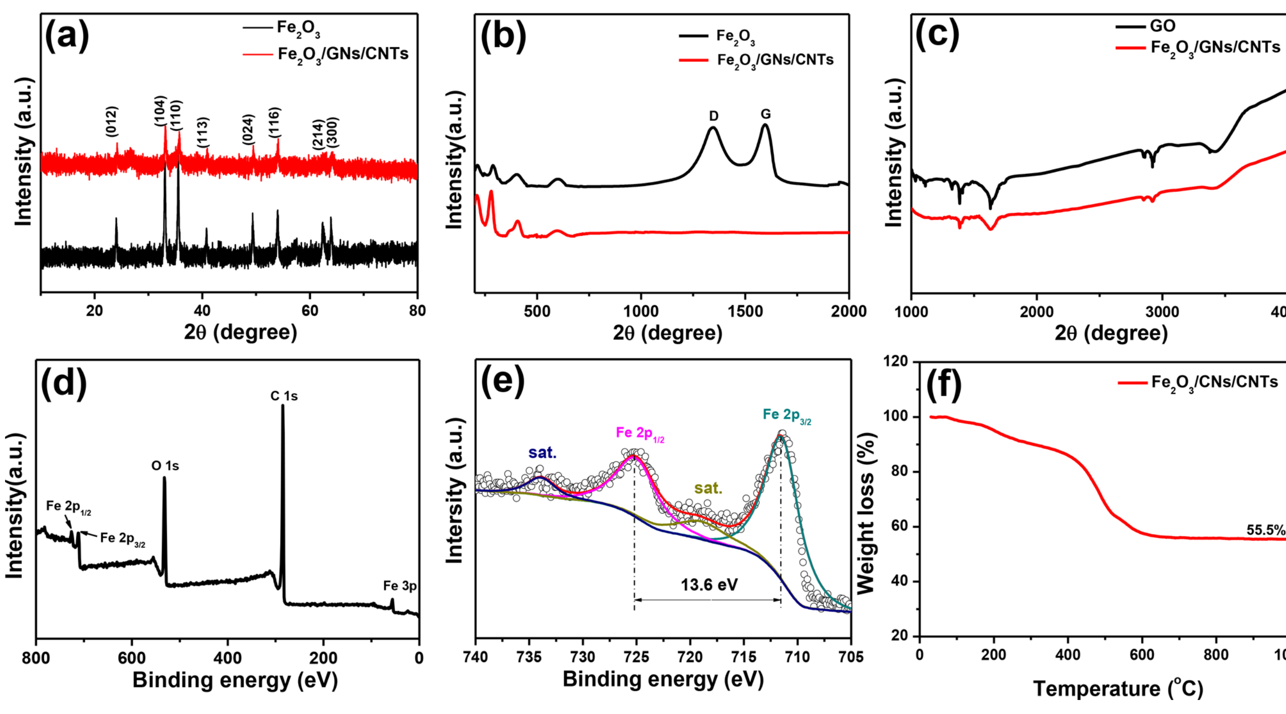


Figure 2. (a) XRD patterns of pure Fe_2O_3 and Fe_2O_3/GNs/CNTs composite. (b) Raman spectra of pure Fe_2O_3 and Fe_2O_3/GNs/CNTs composite. (c) FT-IR spectra of GO and Fe_2O_3/GNs/CNTs composite. (d) XPS survey spectrum of the Fe_2O_3/GNs/CNTs composite. (e) High-resolution Fe 2p spectra of the Fe_2O_3/GNs/CNTs composite. (f) TGA curve of the Fe_2O_3/GNs/CNTs composite.

Especially, the unique structure of graphene can effectively impede the agglomeration and reduce the size of nanoparticles, generating more electroactive sites.[29] Moreover, in contrast to single graphene that enhances merely one aspect of performances, graphene-CNTs conductive networks can improve the overall property of the electrode due to the synergetic effect.[30] Hence, the development of new nanostructured carbon material/Fe_2O_3 is favorable for achieving high performance supercapacitor.

Herein, we develop a convenient route to prepare Fe_2O_3/graphene nanosheets/CNTs (Fe_2O_3/GNs/CNTs) composite as electrode materials for a supercapacitor. Due to the unique electrical and structural support, the Fe_2O_3/GNs/CNTs electrode delivers large specific capacity and superior rate characteristics in 6 M KOH solution. More importantly, the as-

DOI: 10.1021/acssuschemeng.8b06857
ACS Sustainable Chem. Eng. 2019, 7, 9211−9219

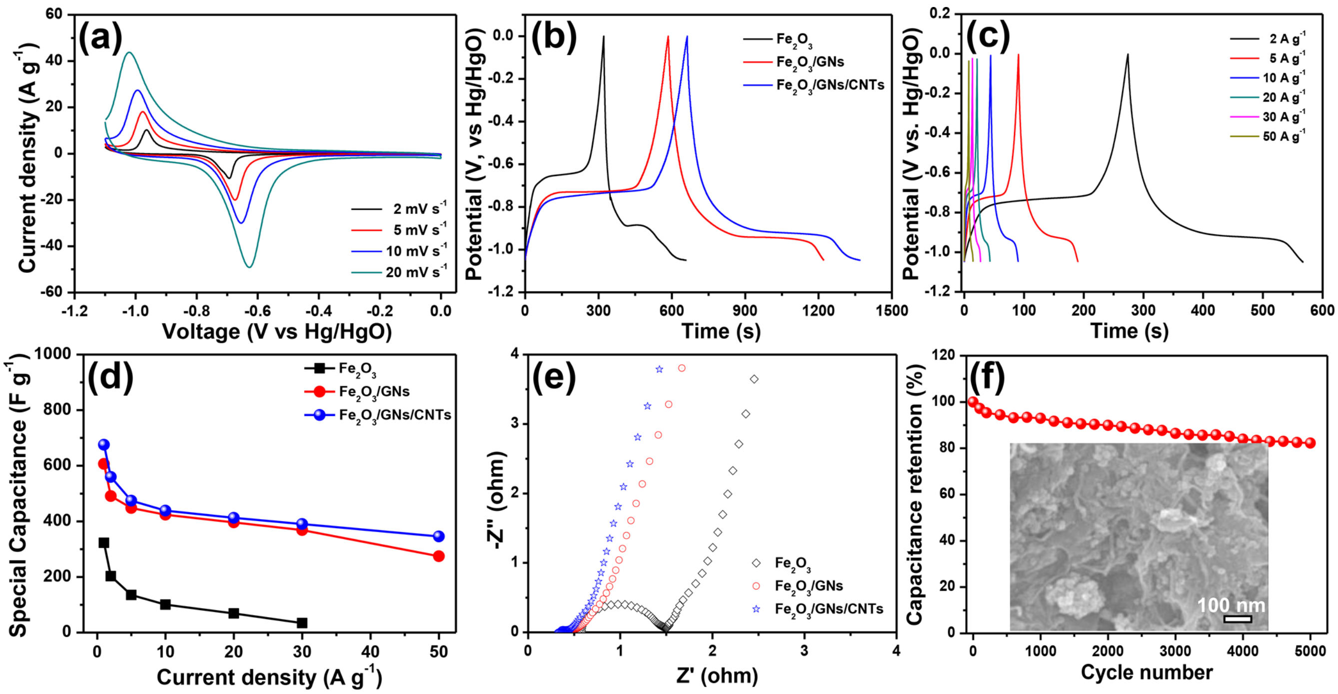


Figure 3. (a) CV curves of Fe_2O_3/GNs/CNTs at different scan rates. (b) Galvanostatic charge−discharge curves of Fe_2O_3, Fe_2O_3/GNs, and Fe_2O_3/GNs/CNTs at a current density of 1 A g^{-1}. (c) Galvanostatic charge−discharge curves of Fe_2O_3/GNs/CNTs at different current densities. (d) Specific capacity of Fe_2O_3, Fe_2O_3/GNs, and Fe_2O_3/GNs/CNTs at different current densities. (e) Nyquist plots of the Fe_2O_3, Fe_2O_3/GNs and Fe_2O_3/GNs/CNTs electrodes. (f) Cycling performance of the Fe_2O_3/GNs/CNTs electrode. The inset illustrates SEM image of Fe_2O_3/GNs/CNTs after 5000 cycle tests.

constructed asymmetric device using Fe_2O_3/GNs/CNTs as the cathode electrode and sulfurized CoAl layered double hydroxides (SLDH) as the anode electrode exhibits high energy density and superior electrochemical steadiness in KOH/PVA gel electrolyte.

RESULTS AND DISCUSSION

Characterization of Cathode Electrode Material. The microstructures of the prepared materials were checked by scanning/transmission electron microscopy (SEM/TEM). As seen in Figure 1a, pure Fe_2O_3 materials display a block-like structure with the size of 20−50 nm. For the Fe_2O_3/GNs composite, the Fe_2O_3 nanoparticles are wrapped by graphene nanosheets (Figure S1a), and the unique structure can effectively prevent the aggregating of Fe_2O_3 and enhanced conductive property of the composite. After the introduction of carbon nanotubes to form a porous but secure structure (Figure 1b), which can offer barrier-free access for electrolyte ions fast diffusion during charge/discharge procedure. The element mapping images of Fe_2O_3/GNs/CNTs exhibit the homogeneous distribution of O, Fe and C elements (Figure S1c-e). Furthermore, the TEM image further confirms that the Fe_2O_3 nanoparticles randomly distribute and closely anchor on the surface of graphene-CNTs conductive networks (Figure 1c), which facilitates the electrons fast transport during the charge/discharge procedure. Remarkably, from the high-resolution TEM observed, the morphology of Fe_2O_3 transforms into ultrasmall nanoparticles (<10 nm, Figure 1d, marked with blue circles) after the introduction of GNs because of more iron ion nucleation sites supplied by GNs.[29]

The crystal structure characteristic of the prepared materials was performed by X-ray diffraction (XRD). As seen in Figure 2a, the diffraction peaks of Fe_2O_3 and Fe_2O_3/GNs/CNTs at about 2θ = 24.0°, 33.1°, 35.5°, 40.7°, 49.4°, 54.0°, 62.5°, and 63.9° can be ascribed to the (012), (104), (110), (113), (024), (116), (214), and (300) planes of Fe_2O_3 (PDF no. 33-0664), respectively. Furthermore, compared with the pure Fe_2O_3, the characteristic peaks of Fe_2O_3/GNs/CNTs became weaker due to the Fe_2O_3 nanoparticles being wrapped by GNs. The peak at 25° represent the (002) plane of GNs cannot be seen for Fe_2O_3/GNs/CNTs due to the surface of graphene-CNTs conductive networks was covered with nanosized Fe_2O_3, and this phenomenon can be found in the reported literatures.[23,30] Raman spectra of the pure Fe_2O_3 and Fe_2O_3/GNs/CNTs composite are displayed in Figure 2b. The diffraction peaks at 214, 288, 403, and 602 cm^{-1} can be found in Fe_2O_3 and Fe_2O_3/GNs/CNTs, which are in good agreement with reported results of Fe_2O_3.[23] In addition, the ratio of D peak to G peak of Fe_2O_3/GNs/CNTs is 0.98, demonstrating partly functional groups remain after reflux. Figure 2c displays the Fourier transform infrared (FT-IR) spectra of graphene oxide (GO) and Fe_2O_3/GNs/CNTs. The absorption bands located at 1643 and 1380 correspond to C═O and C−O in COOH, at 1320 and 1110 cm^{-1} are corresponding to the C−OH stretching vibration, C−O stretching vibration in C−O−C. After reflux, the strength of these peaks for the oxygen functional groups is distinctly weakened, further suggesting the partial removal of oxygen functional groups.

The chemical constitution and bonding states of the Fe_2O_3/GNs/CNTs materials were detected through X-ray photoelectron spectroscopy (XPS). The XPS measurement spectrum of Fe_2O_3/GNs/CNTs confirms that the composite is consisting of C, O, and Fe elements (Figure 2d). The C 1s spectrum (Figure S2a) can be divided into four peaks that ascribed to C = C (284.5 eV), C−C (285.0 eV), C−O (286.7 eV), and C = O (288.6 eV) bonds, respectively, suggesting

DOI: 10.1021/acssuschemeng.8b06857
ACS Sustainable Chem. Eng. 2019, 7, 9211−9219

conductive carbon substrate containing moderate oxygen functional groups. From the Fe 2p spectrum (Figure 2e), two main peaks are situated at 707.4 and 720.1 eV attributed to Fe $2p_{3/2}$ and Fe $2p_{1/2}$, which are in good accordance with the previously reported Fe_2O_3 materials in the literature.[24] The specific surface area of the Fe_2O_3/GNs/CNTs samples were tested through nitrogen adsorption/desorption isotherms (Figure S2b). The Fe_2O_3/GNs/CNTs samples show a large specific surface area of 220 m^2 g^{-1}, larger than the published results in the literatures.[27,28] The mass-loading of Fe_2O_3 in the Fe_2O_3/GNs/CNTs composites was determined by thermal gravimetric analysis (TGA) in air. As seen in Figure 2f, the mass-loading of Fe_2O_3 in the Fe_2O_3/GNs/CNTs samples is about 55.5 wt %.

The electrochemical characteristics of the Fe_2O_3/GNs/CNTs samples were first measured by cyclic voltammetry (CV) in 6 M KOH aqueous electrolyte. As seen in Figure 3a, all of the CV profiles of the Fe_2O_3/GNs/CNTs electrode show one pair of strong redox peaks, demonstrating that the specific capacity primarily comes from Faradaic redox reactions. The strong redox peaks are ascribed to the reversible transition of Fe^{2+} and Fe^{3+}. Remarkably, as the scan rates increase, the contour of CV profiles shows no apparent distortion, indicating fast charge transfer kinetics and superior rate performance. Galvanostatic charge−discharge (GCD) was tested to assess the electrochemical behavior of the Fe_2O_3/GNs/CNTs materials. The GCD profiles of the Fe_2O_3, Fe_2O_3/GNs, and Fe_2O_3/GNs/CNTs electrodes show obvious Faraday pseudocapacity characteristic with a discharge platform occur at around −0.9 V (Figure 3b). The Fe_2O_3/GNs/CNTs electrode delivers a longer discharge time than other electrodes, suggesting a higher specific capacitance for Fe_2O_3/GNs/CNTs. The specific capacities of the Fe_2O_3/GNs/CNTs electrodes were calculated and displayed in Figure 3d based on the discharge curves (Figure 3c). The Fe_2O_3/GNs/CNTs electrode delivers a high specific capacitance of 675.7 F g^{-1} at 1 A g^{-1} based on the electroactive material, which is comparable with those of Fe_2O_3 (322.6 F g^{-1}), Fe_2O_3/GNs (606.5 F g^{-1}), and other formerly published Fe-based electrodes in literatures (Table 1). Remarkably, even the Fe_2O_3/GNs/CNTs electrode even retains 345.4 F g^{-1} at 50 A g^{-1}, meaning superior rate characteristic. Electrochemical impedance was further investigated the electrode kinetics of the obtained samples. As seen in Figure 3e, the ohmic resistance of the electrolyte and cell components ($R(e)$) of the Fe_2O_3/GNs/CNTs electrode (0.31 Ω) is lower than Fe_2O_3/GNs (0.39 Ω) and Fe_2O_3 (0.59 Ω), indicating a much better conductivity for Fe_2O_3/GNs/CNTs. Moreover, the combination of surface and charge-transfer resistance $R(s + ct)$ of the Fe_2O_3/GNs/CNTs electrode is much lower than Fe_2O_3 and Fe_2O_3/GNs, confirming a more rapid charge-transfer rate for Fe_2O_3/GNs/CNTs. Cycling life of the Fe_2O_3/GNs/CNTs electrode was measured at 5 A g^{-1} for 5000 cycles. As displayed in Figure 3f, after 5000 cycle tests, the Fe_2O_3/GNs/CNTs electrode can keep 82.4% of the original capacity, indicating superior cycling steadiness. Notably, after the cycling process, the Fe_2O_3/GNs/CNTs samples can still maintain the original stable structure with almost no deformation.

Table 1. Summary of Electrochemical Performance for Fe_2O_3-Based Electrode Materials

material	C (F g^{-1})	electrolyte	ref.
Fe_2O_3	100.6 (1 mA cm^{-2})	3 M LiCl	9
Fe_2O_3	908.0 (2 A g^{-1})	2 M KOH	31
Fe_2O_3	558.7 F g^{-1} (1 A g^{-1})	2 M KOH	32
Fe_2O_3	257.8 (1.4 A g^{-1})	5 M LiCl	33
Fe_2O_3	116.3 (5 mV s^{-1})	1 M Li_2SO_4	34
Fe_2O_3/graphene	618.0 (0.5 A g^{-1})	1 M KOH	27
Fe_2O_3/graphene	908.0 (2 A g^{-1})	1 M KOH	28
Fe_2O_3/carbon black	40.1 (10 mV s^{-1})	2 M KCl	35
Fe_2O_3 /graphene	215.0 (2.5 mV s^{-1})	1 M Na_2SO_4	36
Fe_2O_3/graphene	343.7 (3 A g^{-1})	1 M Na_2SO_4	37
Fe_2O_3/graphene	226.0 (1 A g^{-1})	1 M Na_2SO_4	38
Fe_2O_3/graphene	224.0 (25 mV s^{-1})	1 M Na_2SO_3	39
Fe_2O_3/graphene	151.8 (1 A g^{-1})	2 M KOH	40
Fe_2O_3/graphene	504 (2 mA cm^{-2})	1 M Na_2SO_4	41
Fe_2O_3/graphene	306.9 (3 A g^{-1})	1 M Na_2SO_4	42
Fe_2O_3/GNs/CNTs	675.7 (1 A g^{-1})	6 M KOH	this work

The Fe_2O_3/GNs/CNTs composite shows good electrochemical properties because of the unique structure. (1) The large specific surface area of Fe_2O_3/GNs/CNTs can supply massive reactive sites to sufficiently utilize the pseudocapacitive characteristics of Fe_2O_3. (2) Fe_2O_3 nanoparticles are wrapped by graphene nanosheets with CNTs-bridged frameworks to form a porous but secure structure, which can not only offer barrier-free access for electrolyte ions fast transport during charge/discharge procedure but also effectively buffer volume change of Fe_2O_3/GNs/CNTs during the charge/discharge procedure to ensure good electrochemical stability. (3) The outstanding conductivity of graphene and CNTs can construct a fast conductive network for electrons rapid transport along overall of the electrode materials.

Characterization of Anode Electrode Materials. The morphology and microstructure of the CoAl layered double hydroxides (LDH) and SLDH samples were checked by SEM and TEM. As seen in Figure 4a, the SEM image of LDH materials show a lamellar structure and agglomerated together. After sulfurization, the SLDH samples (Figure 4b) still retained a lamellar structure with part of the fragmentation. Furthermore, the TEM image of LDH materials (Figure 4c) confirms the lamellar structure with the lateral dimension in the scope of 30−60 nm. After sulfurization, the SLDH samples (Figure 4d) display an amorphous lamellar porous structure, which can not only generate more redox active sites but also facilitate the ions rapid diffusion. The crystal structures of the LDH and SLDH samples were checked by XRD measurements. The XRD pattern (Figure 5a) of the LDH samples is in accord with CoAl-LDH (JCPDS No. 51-0045). Nevertheless, no distinct characteristic peaks can be detected in the XRD pattern of SLDH, suggesting an amorphous structure after the sulfurization, which is in agreement with the published result in the literature.[43] The absence of long-range crystalline order in the process of sulfur substituted hydroxide and, therefore, the amorphous structure. The chemical constitution and bonding states of the SLDH material were checked through X-ray photoelectron spectroscopy (XPS). As seen in Figure 5b, the XPS survey spectrum of SLDH reveals the composites consists of Co, Al, O, and S elements. The Co 2p spectrum (Figure 5c) was spin−orbit split into two major peaks situated at 781.2 and 797.1 eV with satellites accordingly, which are composed of Co $2p_{1/2}$ and Co $2p_{3/2}$ components, demonstrating the existence of Co^{2+} and Co^{3+} in SLDH samples. For S 2p spectrum (Figure 5d), the peaks at around 163.2 and 162.0 eV were

DOI: 10.1021/acssuschemeng.8b06857
ACS Sustainable Chem. Eng. 2019, 7, 9211−9219

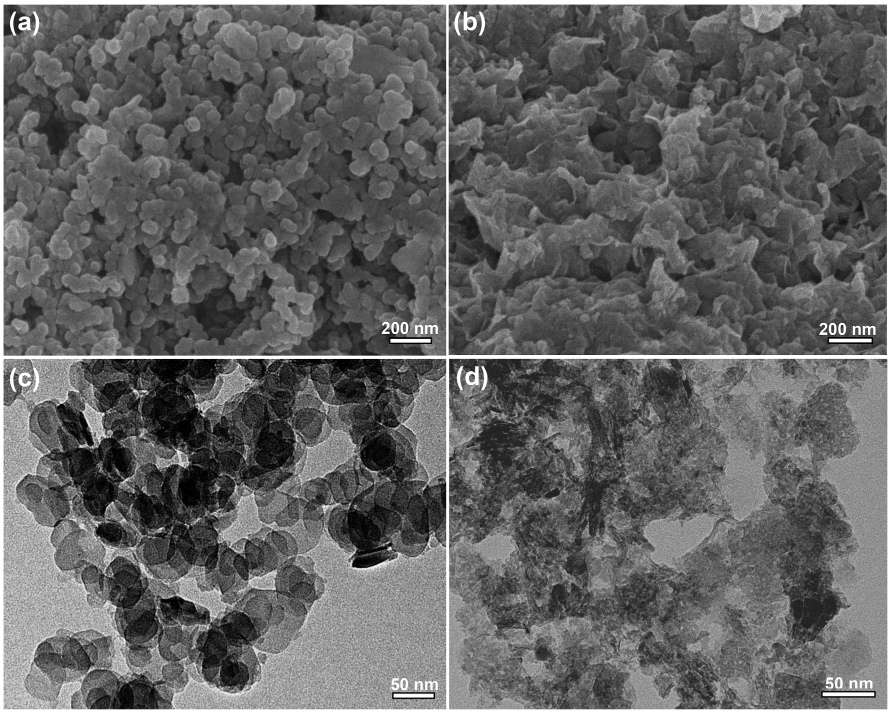


Figure 4. (a) SEM image of LDH. (b) SEM image of SLDH. (c) TEM image of LDH. (d) TEM image of SLDH.

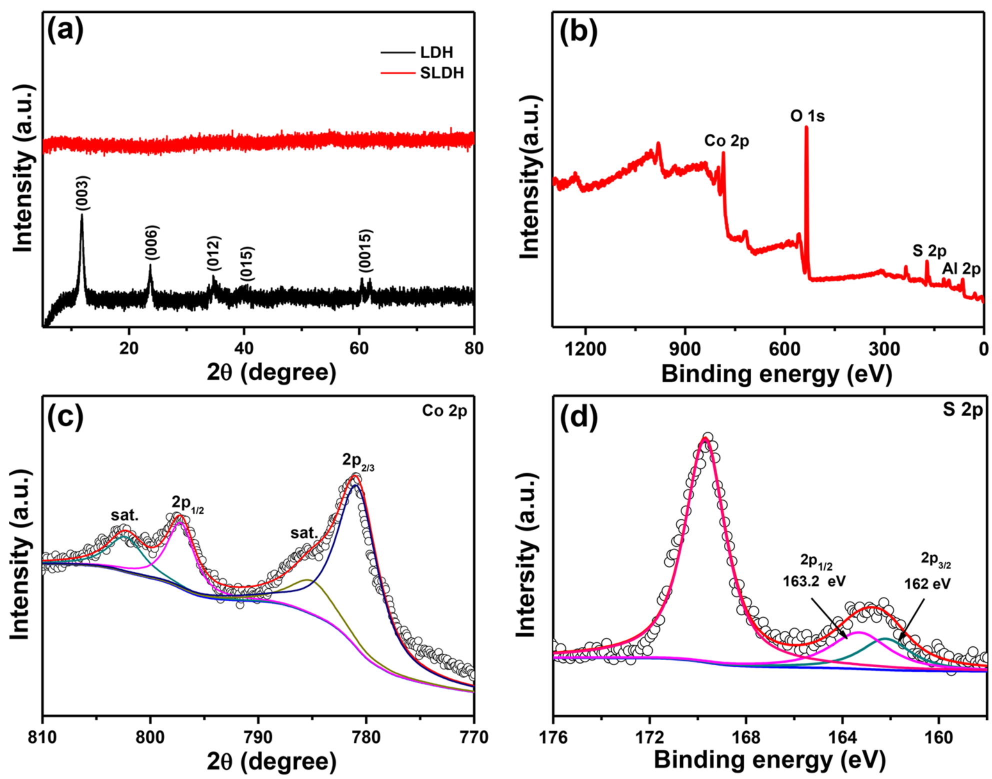


Figure 5. (a) XRD patterns of the LDH and SLDH samples. (b) XPS survey spectrum of the SLDH samples. (c) High-resolution Co 2p spectra of the SLDH samples. (d) High-resolution S 2p spectra of the SLDH samples.

attributed to S 2p1/2 and S 2p3/2 of metal−sulfur bonds. The peak at 69.86 eV was attributed to Al 2p (Figure S3a), confirming the presence of Al^{3+} in the composite. The nitrogen adsorption/desorption isotherms of SLDH (Figure S3b) show a typical IV isotherm with a high specific surface area of 58 m^2 g^{-1} due to the porous structure after sulfurization.

The electrochemical capabilities of the SLDH samples were measured by CV and GCD tests in 6 M KOH solution. The CV profiles of the LDH and SLDH electrode show one pair of obvious redox peaks at 10 mV s^{-1} (Figure 6a), suggesting the Faradaic pseudocapacitive characteristics of the LDH and SLDH electrodes. Furthermore, it can be clearly observed that

DOI: 10.1021/acssuschemeng.8b06857
ACS Sustainable Chem. Eng. 2019, 7, 9211−9219

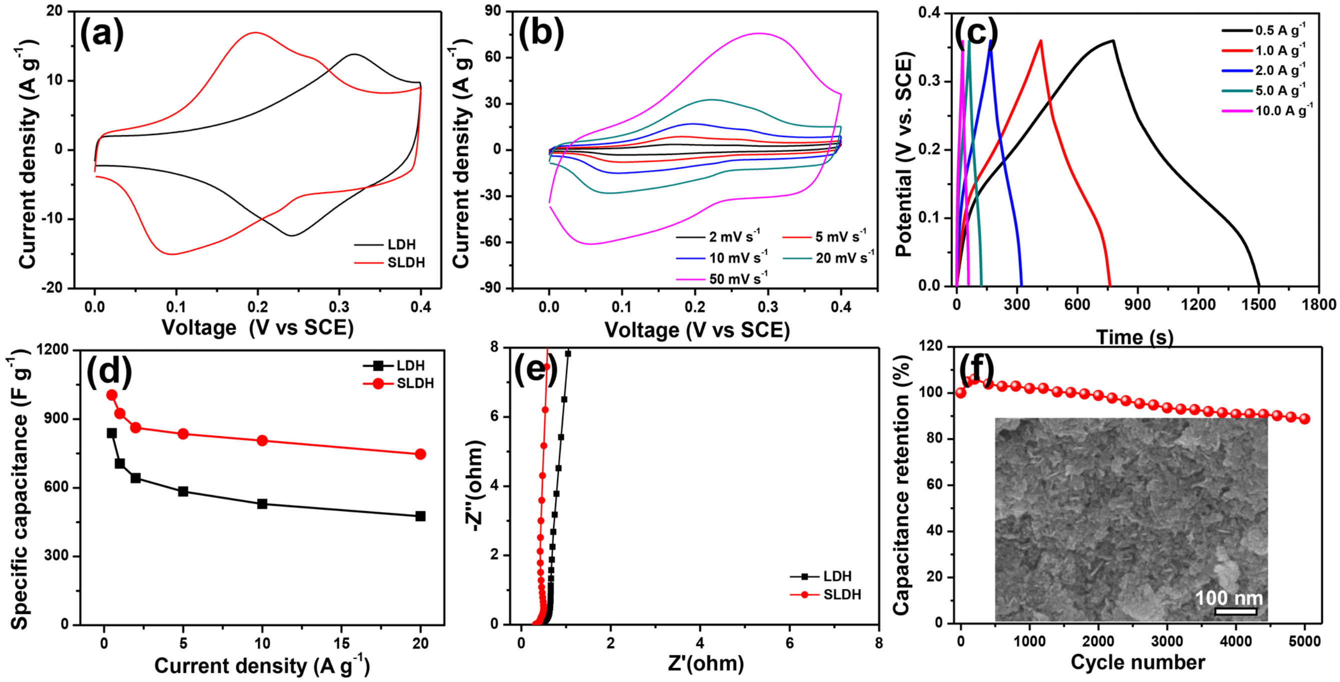


Figure 6. (a) CV curves of LDH and SLDH at the scan rate of 10 mV s^{-1}. (b) CV curves of S-LDH at different scan rates. (c) Galvanostatic charge−discharge profiles of SLDH at different current densities. (d) Specific capacity of LDH and SLDH at different current densities. (e) Nyquist plots of the LDH and SLDH electrodes. (f) Cycling performance of the SLDH electrode. The inset illustrates SEM image of SLDH after 5000 cycle tests.

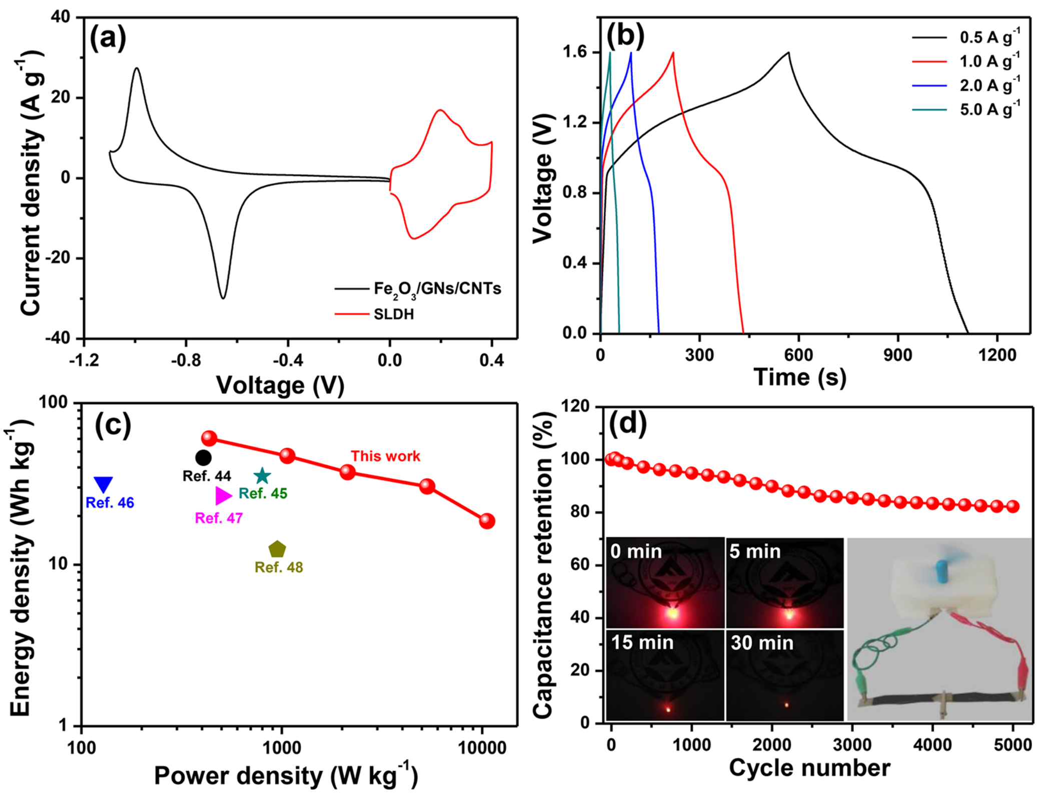


Figure 7. (a) Comparative CV curves of Fe_2O_3/GNs/CNTs and SLDH electrodes performed in a three-electrode cell in 6 M KOH aqueous solution at a scan rate of 10 mV s^{-1}. (b) Galvanostatic charge−discharge curves of the Fe_2O_3/GNs/CNTs//SLDH asymmetric supercapacitor at different scan rates with PVA/KOH gel electrolyte. (c) Ragone plot of the Fe_2O_3/GNs/CNTs//SLDH all-solid-state asymmetric supercapacitor and the previously reported values. (d) Cycling performance of the Fe_2O_3/GNs/CNTs//SLDH all-solid-state asymmetric supercapacitor.

DOI: 10.1021/acssuschemeng.8b06857
ACS Sustainable Chem. Eng. 2019, 7, 9211−9219

the redox peak shifts to the cathode range after sulfur substitution. The possible Faraday redox reaction for SLDH electrode could be expressed by the following expression:

$$CoS + OH^- \rightarrow CoSOH + e^- \quad (1)$$

$$CoSOH + OH^- \rightarrow CoSO + H_2O + e^- \quad (2)$$

The CV curve of the SLDH electrode shows a relative bigger integrated area than that of LDH electrode, demonstrating a higher specific capacity for SLDH. The increased specific capacity is due to the amorphous porous lamellar structure after sulfurization, which can supply more redox active sites and facilitate ion fast diffusion during charge−discharge procedure. As displayed in Figure 6b, even at large scan rates, the CV profile of the SLDH electrode was well-maintained, indicating fast charge transfer kinetics and good rate characteristic. Moreover, in accord with CV results, the GCD profiles of the SLDH electrode shows typical pseudocapacitive characteristics (Figure 6c). The GCD profile of the SLDH electrode shows no distinct IR drop at 10 A g^{-1}, suggesting small internal resistance. The specific capacities were computed according to the GCD profiles and shown in Figure 6d. The SLDH electrode displays a superior specific capacitance of 1005.1 F g^{-1} at 0.5 A g^{-1} according to the electroactive materials, much higher than that of LDH. Significantly, the SLDH electrode shows a good rate performance of 74.3% at 20 A g^{-1}, much better than that of LDH and previously reported LDH and metal sulfide in the literatures (Table S1). The good rate performance is due to the amorphous porous structure after sulfurization, which is in favor of the rapid diffusion of ions. To explore the electrode kinetics, electrochemical impedance spectroscopy was tested and shown in Figure 6e. The $R(e)$ of the SLDH electrode is 0.32 Ω, which is lower than that of LDH (0.36 Ω), indicating a much better conductivity for SLDH. Additionally, in the low-frequency scope, the SLDH electrode displays an almost vertical line, meaning a fast electron transfer rate. Cycling life of the SLDH electrode was measured at 5 A g^{-1} and displayed in Figure 6f. The SLDH electrode can keep 88.7% of the original capacity after 5000 cycles, confirming superior cycling steadiness. Notably, after cycling process, the SLDH samples can still maintain the original lamellar structure with almost no deformation.

Characterizations of Asymmetric Supercapacitor. To further assess the electrochemical characteristics of the obtained samples, an all-solid-state ASC equipment was assembled using the Fe_2O_3/GNs/CNTs material as the cathode electrode and the SLDH material as the anode electrode in PVA/KOH gel electrolyte. Figure 7a shows a steady potential region range from −1.1 to 0 V for Fe_2O_3/GNs/CNTs electrode and from 0.0 to 0.4 V for SLDH electrode in a three-electrode system. The quality ratio of the anode electrode to cathode electrode was computed based on formula (S3). The CV profiles of the optimized Fe_2O_3/GNs/CNTs//SLDH ASC were measured at 10 mV s^{-1} in different potential region in the PVA/KOH gel electrolyte. As seen in Figure S4a, no distinct increase of anodic current can be seen even at 1.6 V, indicating a wide potential region of 0−1.6 V can be steadily obtained. The wide voltage range is conducive to achieving a larger energy density; hence, the electrochemical property of the ASC were tested from 0 to 1.6 V. Figure S4b displays the CV profiles of the optimized ASC measured at various scan rates in the voltage range of 0−1.6 V. The contour of the CV profiles demonstrates that the capacity originates from the combination of both pseudocapacity and electric double-layer capacity. The GCD profiles of the ASC shows typical pseudocapacitive characteristics (Figure 7b). The all-solid-state ASC has a large specific capacity of 169.5 F g^{-1} at 0.5 A g^{-1}. Benefiting from its wide voltage range and high specific capacity, the optimized ASC achieves a high energy density of 60.3 Wh kg^{-1} (Figure 7c), which is better than these of previous published Fe-based ASC devices, such as, Fe_2O_3/CNT//MnO_2/CNT (45.8 Wh kg^{-1}),[43] Fe_3O_4//Co_2AlO_4@MnO_2 (35.25 Wh kg^{-1}),[44] Fe_2O_3//V_2O_5 (32.2 Wh kg^{-1}),[45] Fe_3O_4@Fe_2O_3//Fe_3O_4@MnO_2 (26.6 Wh kg^{-1}),[46] and Fe_2O_3//NiO (12.4 Wh kg^{-1}).[47,48] Figure S4c displays the CV profiles of the assembled ASC at different bent angles. In addition, the Fe_2O_3/GNs/CNTs//SLDH ASC can maintain an initial specific capacity of 82.2% after cyclic tests of 5000 cycles at 5 A g^{-1} (Figure 7d), demonstrating good electrochemical stability. More interestingly, two series connected Fe_2O_3/GNs/CNTs//SLDH ASC devices can light the red commercial light emitting diodes (LEDs) for 30 min and run a small fan, demonstrating its practical application (inset of Figure 7d).

■ CONCLUSIONS

In summary, we develop a convenient, one-step route to prepare Fe_2O_3/graphene nanosheets/CNTs composite as electrode materials for supercapacitor. Benefiting from the unique structure, the Fe_2O_3/GNs/CNTs electrode delivers superior specific capacity and superior rate characteristic in 6 M KOH solution. More importantly, the as-constructed all-solid-state ASC using Fe_2O_3/GNs/CNTs as cathode electrode and SLDH as anode electrode shows high energy density and excellent electrochemical steadiness in KOH/PVA gel electrolyte.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acssuschemeng.8b06857.

Experimental section, SEM image, elemental mapping images, XPS spectra, nitrogen adsorption−desorption isotherms, CV curves, and specific capacitance at different scan rates (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: wuxiaoliang90@163.com.

ORCID

Xiaoliang Wu: 0000-0002-9297-0922

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by National Natural Science Foundation of China (51702043) and Heilongjiang Postdoctoral Foundation (LBH-Z18008).

REFERENCES

(1) Guan, B.; Yu, X.; Wu, H.; Lou, X. Complex nanostructures from materials based on metal−organic frameworks for electrochemical energy storage and conversion. *Adv. Mater.* **2017**, *29*, 1703614.

(2) Wang, Q.; Yan, J.; Fan, Z. Carbon materials for high volumetric performance supercapacitors: design, progress, challenges and opportunities. *Energy Environ. Sci.* **2016**, *9*, 729−762.

(3) Zardkhoshoui, A. M.; Davarani, S. S. H. Flexible asymmetric supercapacitors based on CuO@MnO_2-rGO and MoS_2-rGO with ultrahigh energy density. *J. Electroanal. Chem.* **2018**, *827*, 221−229.

(4) Ling, Z.; Wang, Z. Y.; Zhang, M. D.; Yu, C.; Wang, G.; Dong, Y. F.; Liu, S. H.; Wang, Y. W.; Qiu, J. S. Sustainable synthesis and assembly of biomass-derived B/N co-doped carbon nanosheets with ultrahigh aspect ratio for high-performance supercapacitors. *Adv. Funct. Mater.* **2016**, *26*, 111−119.

(5) Ding, B.; Guo, D.; Wang, Y. H.; Wu, X. L.; Fan, Z. J. Functionalized graphene nanosheets decorated on carbon nanotubes networks for high performance supercapacitors. *J. Power Sources* **2018**, *398*, 113−119.

(6) Guo, D.; Ding, B.; Hu, X.; Wang, Y.; Han, F.; Wu, X. Synthesis of boron and nitrogen codoped porous carbon foam for high performance supercapacitors. *ACS Sustainable Chem. Eng.* **2018**, *6*, 11441−11449.

(7) Zardkhoshoui, A. M.; Davarani, S. S. H. Synthesis of $NiGa_2S_4$-rGO on nickel foam as advanced electrode for flexible solid-state supercapacitor with superior energy density. *J. Colloid Interface Sci.* **2019**, *535*, 195−204.

(8) Wu, X.; Jiang, L.; Long, C.; Wei, T.; Fan, Z. Dual support system ensuring porous Co−Al hydroxide nanosheets with ultrahigh rate performance and high energy density for supercapacitors. *Adv. Funct. Mater.* **2015**, *25*, 1648−1655.

(9) Yang, J.; Yu, C.; Fan, X. M.; Liang, S. X.; Li, S. F.; Huang, H. W.; Ling, Z.; Hao, C.; Qiu, J. S. Electroactive edge site-enriched nickel-cobalt sulfide into graphene frameworks for high-performance asymmetric supercapacitors. *Energy Environ. Sci.* **2016**, *9*, 1299−1307.

(10) Lei, Z.; Zhang, J.; Zhao, X. Ultrathin MnO_2 nanofibers grown on graphitic carbonspheres as high-performance asymmetric supercapacitor electrodes. *J. Mater. Chem.* **2012**, *22*, 153−160.

(11) Yan, J.; Fan, Z.; Sun, W.; Ning, G.; Wei, T.; Zhang, Q.; Zhang, R.; Zhi, L.; Wei, F. Advanced asymmetric supercapacitors based on $Ni(OH)_2$/graphene and porous graphene electrodes with high energy density. *Adv. Funct. Mater.* **2012**, *22*, 2632−2641.

(12) Zardkhoshoui, A. M.; Davarani, S. S. H.; Hashemi, M. Fabrication of cobalt gallium oxide with zinc iron oxide on nickel foam for a high-performance asymmetric supercapacitor. *New J. Chem.* **2019**, *43*, 4590−4598.

(13) Chang, J.; Jin, M.; Yao, F.; Kim, T.; Le, V.; Yue, H.; Gunes, F.; Li, B.; Ghosh, A.; Xie, S.; Lee, Y. Asymmetric supercapacitors based on graphene/MnO_2 nanospheres and graphene/MoO_3 nanosheets with high energy density. *Adv. Funct. Mater.* **2013**, *23*, 5074−5083.

(14) Xu, J.; Wang, Q. F.; Wang, X. M.; Xiang, Q. Y.; Liang, B.; Chen, D.; Shen, G. Z. Flexible asymmetric supercapacitors based on Co_9S_8 nanorod//Co_3O_4@RuO_2 nanosheet arrays on carbon cloth. *ACS Nano* **2013**, *7*, 5453−5462.

(15) Zardkhoshoui, A. M.; Davarani, S. S. H. Ultrahigh energy density supercapacitors based on facile synthesized Ni,CoOH-rGO/NF hybrid electrodes. *J. Alloys Compd.* **2018**, *769*, 922−931.

(16) Yang, J.; Yu, C.; Fan, X.; Qiu, J. 3D architecture materials made of NiCoAl-LDH nanoplates coupled with NiCo-carbonate hydroxide nanowires grown on flexible graphite paper for asymmetric supercapacitors. *Adv. Energy Mater.* **2014**, *4*, 1400761.

(17) Zardkhoshoui, A. M.; Davarani, S. S. H. All-solid-state, flexible, ultrahigh performance supercapacitors based on the Ni-Al LDH-rGO electrodes. *J. Alloys Compd.* **2018**, *750*, 515−522.

(18) Li, Y. H.; Cao, L. J.; Qiao, L.; Zhou, M.; Yang, Y.; Xiao, P.; Zhang, Y. H. Ni-Co sulfide nanowires on nickel foam with ultrahigh capacity for asymmetric supercapacitors. *J. Mater. Chem. A* **2014**, *2*, 6540−6548.

(19) Zeng, Y. X.; Yu, M. H.; Meng, Y.; Fang, P. P.; Lu, X. H.; Tong, Y. X. Iron-based supercapacitor electrodes: advances and challenges. *Adv. Energy Mater.* **2016**, *6*, 1601053.

(20) Nithya, V. D.; Arul, N. S. Review on a-Fe_2O_3 based cathode electrode for high performance supercapacitors. *J. Power Sources* **2016**, *327*, 297−318.

(21) Zardkhoshoui, A. M.; Davarani, S. S. H. Designing a flexible all-solid-state supercapacitor based on $CuGa_2O_4$ and FeP-rGO electrodes. *J. Alloys Compd.* **2019**, *773*, 527−536.

(22) Liang, H.; Xia, C.; Emwas, A.; Anjum, D.; Miao, X.; Alshareef, H. Phosphine plasma activation of α-Fe_2O_3 for high energy asymmetric supercapacitors. *Nano Energy* **2018**, *49*, 155−162.

(23) Wang, Y.; Zhang, M.; Pan, D.; Li, Y.; Ma, T.; Xie, J. Nitrogen/sulfur co-doped graphene networks uniformly coupled N-Fe_2O_3 nanoparticles achieving enhanced supercapacitor performance. *Electrochim. Acta* **2018**, *266*, 242−253.

(24) Chen, J.; Xu, J.; Zhou, S.; Zhao, N.; Wong, C. Template-grown graphene/porous Fe_2O_3 nanocomposite: A high-performance anode material for pseudocapacitors. *Nano Energy* **2015**, *15*, 719−728.

(25) Shivakumara, S.; Penki, T. R.; Munichandraiah, N. Synthesis and characterization of porous flowerlike α-Fe_2O_3 nanostructures for supercapacitor application. *ECS Electrochem. Lett.* **2013**, *2*, 60−62.

(26) Zheng, X.; Yan, X. Q.; Sun, Y. H.; Yu, Y. S.; Zhang, G. J.; Shen, Y. W.; Liang, Q. J.; Liao, Q. L.; Zhang, Y. Temperature-dependent electrochemical capacitive performance of the α-Fe_2O_3 hollow nanoshuttles as supercapacitor electrodes. *J. Colloid Interface Sci.* **2016**, *466*, 291−296.

(27) Ma, Z.; Huang, X.; Dou, S.; Wu, J.; Wang, S. One-pot synthesis of Fe_2O_3 nanoparticles on nitrogen-doped graphene as advanced supercapacitor electrode materials. *J. Phys. Chem. C* **2014**, *118*, 17231−17239.

(28) Wang, H.; Xu, Z.; Yi, H.; Wei, H.; Guo, Z.; Wang, X. One-step preparation of single-crystalline Fe_2O_3 particles/graphene composite hydrogels as high performance anode materials for supercapacitors. *Nano Energy* **2014**, *7*, 86−96.

(29) Yan, J.; Fan, Z.; Sun, W.; Ning, G.; Wei, T.; Zhang, Q.; Zhang, R.; Zhi, L.; Wei, F. Advanced asymmetric supercapacitors based on $Ni(OH)_2$/graphene and porous graphene electrodes with high energy density. *Adv. Funct. Mater.* **2012**, *22*, 2632−2641.

(30) Long, C.; Jiang, L.; Wei, T.; Yan, J.; Fan, Z. High-performance asymmetric supercapacitors with lithium intercalation reaction using metal oxide-based composites as electrode materials. *J. Mater. Chem. A* **2014**, *2*, 16678−16686.

(31) Tang, Q.; Wang, W.; Wang, G. The perfect matching between the low-cost Fe_2O_3 nanowire anode and the NiO nanoflake cathode significantly enhances the energy density of asymmetric supercapacitors. *J. Mater. Chem. A* **2015**, *3*, 6662−6670.

(32) Zhang, M.; Chen, K.; Chen, X.; Peng, X.; Sun, X.; Xue, D. Ethylenediamine-assisted crystallization of Fe_2O_3 microspindles with controllable size and their pseudocapacity performance. *CrystEngComm* **2015**, *17*, 1521−1525.

(33) Yang, P.; Ding, Y.; Lin, Z.; Chen, Z.; Li, Y.; Qiang, P.; Ebrahimi, M.; Mai, W.; Wong, C.; Wang, Z. Low-cost high-performance solid-state asymmetric supercapacitors based on MnO_2 nanowires and Fe_2O_3 nanotubes. *Nano Lett.* **2014**, *14*, 731−736.

(34) Wang, D.; Wang, Q.; Wang, T. Controlled synthesis of mesoporous hematite nanostructures and their application as electrochemical capacitor electrodes. *Nanotechnology* **2011**, *22*, 135604.

(35) Nasibi, M.; Golozar, M.; Rashed, G. Nano iron oxide (Fe_2O_3)/carbon black electrodes for electrochemical capacitors. *Mater. Lett.* **2012**, *85*, 40−43.

(36) Lee, K.; Deng, S.; Fan, H.; Mhaisalkar, S.; Tan, H.; Tok, E.; Loh, K.; Chin, W.; Sow, C. α-Fe_2O_3 nanotubes-reduced graphene oxide composites as synergistic electrochemical capacitor materials. *Nanoscale* **2012**, *4*, 2958−2961.

(37) Yang, S.; Song, X.; Zhang, P.; Gao, L. Heating-rate-induced porous α-Fe_2O_3 with controllable pore size and crystallinity grown on

graphene for supercapacitors. *ACS Appl. Mater. Interfaces* **2015**, *7*, 75−79.

(38) Wang, Z.; Ma, C.; Wang, H.; Liu, Z.; Hao, Z. Facilely synthesized Fe_2O_3−graphene nanocomposite as novel electrode materials for supercapacitors with high performance. *J. Alloys Compd.* **2013**, *552*, 486−491.

(39) Chen, H.; Wang, C.; Lu, S. γ-Fe_2O_3/graphene nanocomposites as a stable high performance anode material for neutral aqueous supercapacitors. *J. Mater. Chem. A* **2014**, *2*, 16955−16962.

(40) Wang, D.; Li, Y.; Wang, Q.; Wang, T. Nanostructured Fe_2O_3−graphene composite as a novel electrode material for supercapacitors. *J. Solid State Electrochem.* **2012**, *16*, 2095−2102.

(41) Low, Q.; Ho, G. Facile structural tuning and compositing of iron oxide-graphene anode towards enhanced supacapacitive performance. *Nano Energy* **2014**, *5*, 28−35.

(42) Yang, S.; Song, X.; Zhang, P.; Sun, J.; Gao, L. Self-Assembled α-Fe_2O_3 mesocrystals/ graphene nanohybrid for enhanced electrochemical capacitors. *Small* **2014**, *10*, 2270−2279.

(43) Wang, H.; Tang, C.; Wang, B.; Li, B.; Zhang, Q. Bifunctional transition metal hydroxysulfides: room-temperature sulfurization and their applications in Zn−Air batteries. *Adv. Mater.* **2017**, *29*, 1702327.

(44) Gu, T.; Wei, B. High-performance all-solid-state asymmetric stretchable supercapacitors based on wrinkled MnO_2/CNT and Fe_2O_3/CNT macrofilms. *J. Mater. Chem. A* **2016**, *4*, 12289−12295.

(45) Li, F.; Chen, H.; Liu, X.; Zhu, S.; Jia, J.; Xu, C.; Dong, F.; Wen, Z.; Zhang, Y. Low-cost high-performance asymmetric supercapacitors based on Co_2AlO_4@MnO_2 nanosheets and Fe_3O_4 nanoflakes. *J. Mater. Chem. A* **2016**, *4*, 2096−2104.

(46) Jiang, H.; Niu, H.; Yang, X.; Sun, Z.; Li, F.; Wang, Q.; Qu, F. Flexible Fe_2O_3 and V_2O_5 nanofibers as binder-free electrodes for high-performance all-solid-state asymmetric supercapacitors. *Chem. - Eur. J.* **2018**, *24*, 10683−10688.

(47) Tang, X.; Jia, R.; Zhai, T.; Xia, H. Hierarchical Fe_3O_4@Fe_2O_3 core−shell nanorod arrays as high-performance anodes for asymmetric supercapacitors. *ACS Appl. Mater. Interfaces* **2015**, *7*, 27518−27525.

(48) Zhang, S.; Yin, B.; Wang, Z.; Peter, F. Super long-life all solid-state asymmetric supercapacitor based on NiO nanosheets and α-Fe2O3 nanorods. *Chem. Eng. J.* **2016**, *306*, 193−203.

DOI: 10.1021/acssuschemeng.8b06857
ACS Sustainable Chem. Eng. 2019, 7, 9211−9219